

**DATA STRUCTURES Midterm Exam**

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1. Question: Big-O Analysis

Analyze the 3 given code blocks and determine their Big-O notation.

2. Question: Hashing — Collision Resolution

Given two hash tables and two set of inputs:

a) Insert the first set of inputs into the first table using Separate Chaining for collision resolution. Identify any inputs that could not be inserted.

b) Insert the second set of inputs into the second table using Quadratic Probing for collision resolution. Identify any inputs that could not be inserted.

(Two hash tables given visually, insertion will be visually too.)

3. Question: Quick Sort Analysis (15p)

In the Quick Sort algorithm, assume the pivot is always chosen as the first element. Analyze how the Big-O notation changes for the following two cases:

a) The input list is already sorted.

b) The input list is unsorted.

4. Question: Doubly Linked List

Given the global variable definitions and the main function for a Doubly Linked List, write the complete insert and displayList functions.